MATH INQUIRERS PLACEMENT CHECK

# Ages 11-12

For children finishing Primary 5 • About 30 minutes • No calculator

Name: _____ Date: _____

**How it works.** Your child answers on their own. Read a question aloud if they ask, but do not help with the maths. If they are stuck, say "Try your best, or skip it", and move on. Write the answer in the box. Working on scrap paper is fine.

## Warm-up

1 What is 5/6 - 1/3? Give your answer as a fraction in simplest form.

2 A ribbon 2.4 m long is cut into 6 equal pieces. How long is each piece, in metres?

3 A box holds 24 pencils. How many pencils are in 35 boxes?

4 A rectangle has a perimeter of 36 cm and a length of 11 cm. What is its area in square centimetres?

## Main questions

5 Round 4,386,512 to the nearest hundred thousand.

6 Sam spent 2/5 of his money on a book. He then spent 1/3 of the remainder on lunch. He had $24 left. How much money did he have at first, in dollars?

7 The area of a triangle is 40 square cm. Its base is 16 cm. What is its height in cm?

8 A tank is 50 cm long, 30 cm wide and 40 cm high. A tap fills it at 6 litres per minute. How many minutes to fill it completely? (1 litre = 1000 cubic cm)

9 In a class of 40 pupils, 60% are girls. 25% of the girls wear glasses. How many girls wear glasses?

10 Three angles of a quadrilateral are 90, 110 and 75 degrees. What is the fourth angle, in degrees?

## Brain stretchers

11 Look at the pattern 2, 5, 10, 17, 26, ... What is the 10th number in the pattern?

12 Ali has 3 times as many stickers as Ben. After Ali gives 18 stickers to Ben, Ali has twice as many as Ben. How many stickers did Ali have at first?

FOR THE TUTOR OR PARENT

# Answer key and scoring

Accept equivalent forms (for example $2.05 and 2.05). One mark per question.

## Warm-up (4 marks)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 1 | What is 5/6 - 1/3? Give your answer as a fraction in simplest form. | **1/2** | Look for converting 1/3 to 2/6; common error: subtracting numerators and denominators (4/3) or leaving 3/6. |
| 2 | A ribbon 2.4 m long is cut into 6 equal pieces. How long is each piece, in metres? | **0.4** | Check place value; common error is 0.04 or 4 from misplacing the decimal point. |
| 3 | A box holds 24 pencils. How many pencils are in 35 boxes? | **840** | Watch the multiplication method (24 x 35 = 24 x 30 + 24 x 5); common error: dropping the carried tens. |
| 4 | A rectangle has a perimeter of 36 cm and a length of 11 cm. What is its area in square centimetres? | **77** | Child should find width 7 cm first (36 / 2 - 11). Common error: using 36 - 22 = 14 as the width or answering 7. |

## Main questions (6 marks)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 5 | Round 4,386,512 to the nearest hundred thousand. | **4400000 (4,400,000)** | Look at the ten-thousands digit (8). Common errors: 4,300,000 (truncating) or 4,390,000 (wrong place). |
| 6 | Sam spent 2/5 of his money on a book. He then spent 1/3 of the remainder on lunch. He had $24 left. How much money did he have at first, in dollars? | **60** | Best seen with a bar model: remainder is 3/5, lunch is 1/5 of total, left 2/5 = $24. Common error: treating 1/3 as a fraction of the whole. |
| 7 | The area of a triangle is 40 square cm. Its base is 16 cm. What is its height in cm? | **5** | Child should use area = 1/2 x base x height. Common error: forgetting the 1/2 and answering 2.5, or dividing 40 by 16 only. |
| 8 | A tank is 50 cm long, 30 cm wide and 40 cm high. A tap fills it at 6 litres per minute. How many minutes to fill it completely? (1 litre = 1000 cubic cm) | **10** | Volume 60,000 cubic cm = 60 L. Common errors: not converting to litres, or adding the dimensions. |
| 9 | In a class of 40 pupils, 60% are girls. 25% of the girls wear glasses. How many girls wear glasses? | **6** | Girls = 24, then a quarter = 6. Common error: taking 25% of 40 (10). |
| 10 | Three angles of a quadrilateral are 90, 110 and 75 degrees. What is the fourth angle, in degrees? | **85** | Angle sum of quadrilateral is 360. Common error: using 180. |

## Brain stretchers (2 marks)

| # | Question | Answer | Watch for |
|---|---|---|---|
| 11 | Look at the pattern 2, 5, 10, 17, 26, ... What is the 10th number in the pattern? | **101** | Look for differences 3,5,7,9 or the rule n x n + 1. Common error: guessing a constant difference. |
| 12 | Ali has 3 times as many stickers as Ben. After Ali gives 18 stickers to Ben, Ali has twice as many as Ben. How many stickers did Ali have at first? | **162** | Bar-model or before-after reasoning. Ben starts with 54. Common error: answering Ben's amount (54) or ignoring the change in total. |

YOUR CHILD'S RESULT

# Which level fits?

Add up the marks, then find your row. This is a guide. You make the final choice, and you can change level free of charge before lesson 1.

**Warm-up marks:** _____ / 4 **Main + stretch marks:** _____ / 8

| Main + stretch | What it suggests |
|---|---|
| **6 to 8** | Join **Ages 11-12 (P5)**. Your child has the foundations and will be well challenged. If they scored 8 of 8 and all 4 warm-up marks, you can also try the Ages 12-13 (P6) check. |
| **4 or 5** | Join **Ages 11-12 (P5)**, and expect some support. If the warm-up score is 2 or less, Ages 10-11 (P4) may be the more comfortable start. |
| **0 to 3** | We suggest **Ages 10-11 (P4)**. The ideas build step by step, and a confident start matters more than a fast one. |

**Strong?**
Choose the level that fits the score, not the school year. A child who finds this easy will enjoy the challenge questions in class.

**Not yet?**
That is useful to know, not a failure. Starting one level down means more success and more joy in class.

## Skills covered

Place value and rounding to 10 million; Fraction operations and word problems; Area of triangle; Volume and rate; Percentage; Angles in quadrilaterals; Patterns and heuristics

**Next step.** Send the score and your preferred level to us when you enrol. Prices are the same at every level, and your tutor can talk it through first. Small groups of up to 8, live online.